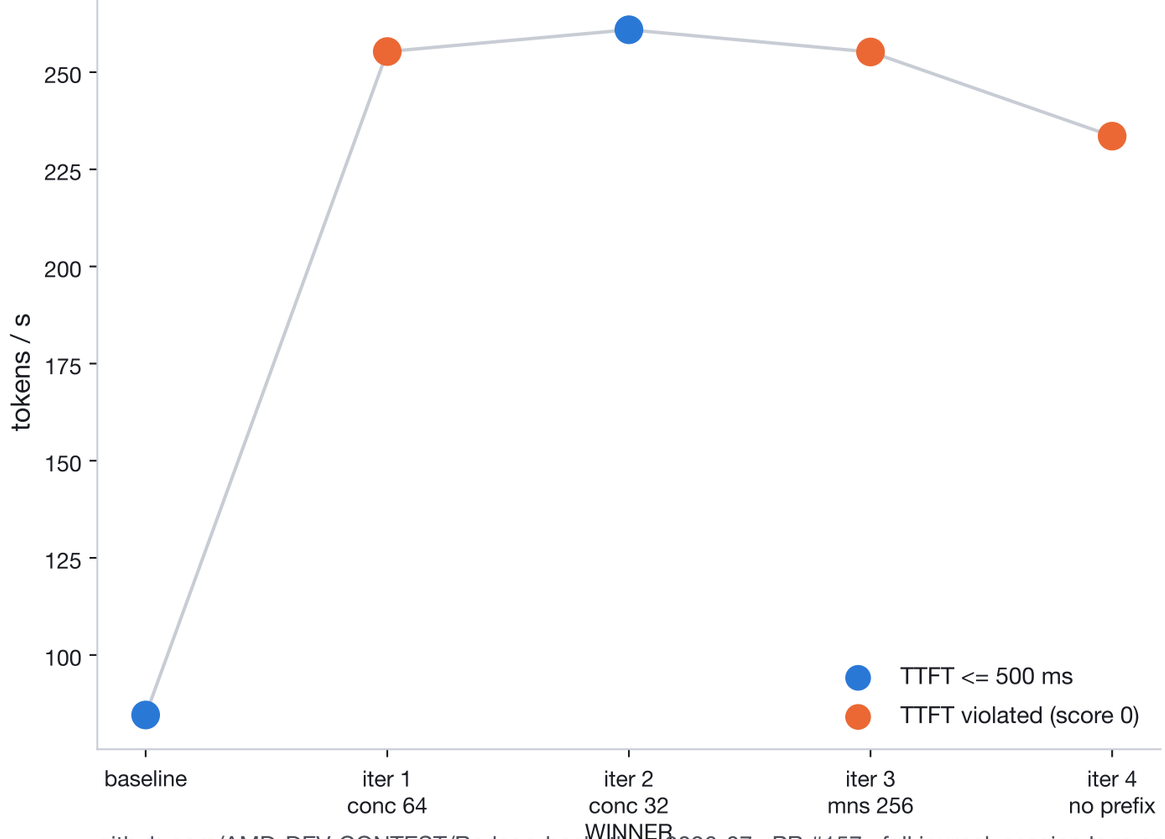
An agent tripled its own throughput on a Radeon GPU and cut energy per token by 68%, in one overnight session

rocm-autopilot · Track 2 Agentic AI · AMD AI DevMaster Hackathon 2026 · Wilfred Doré · Act II of gemma-autopilot (Build with Gemma winner): same agent, new silicon

	vLLM defaults	after the agent	
Throughput	84.6 tok/s	**260.9 tok/s**	x3.08
TTFT (steady state)	388 ms	**367 ms**	guardrail 500 ms held
Energy	2.042 J/token	**0.649 J/token**	-68%
Carbon label	C	**A**	0.10 gCO2e/1k tok

The agent's trajectory: probe hard, get caught by the guardrail, converge



The brain IS the deployment

The model served by vLLM-ROCm reasons about its own benchmark results, acts through bounds-checked tools (max_num_seqs, prefix caching, concurrency, attention backend, power cap), and re-measures until gains flatten. Reasoning, serving, amdsmi telemetry, journal and dashboard all live on one Radeon gfx1100. No remote API anywhere.

Also measured (controls, not vibes)

Cold-start vs steady-state isolated: TTFT 990 ms vs 367 ms, the win is prefix-cache exploitation and every claim is labeled accordingly. Attention-backend sweep: TRITON / ROCM / FLASH / AITER_UNIFIED all run on gfx1100 within 0.4% throughput; ROCM_ATTN most frugal.

Honest limitations are part of the product

The SKU's energy counter is dead: energy is measured by integrating amdsmi socket power at 2 Hz; every run records method=power_integration. Power capping is denied in the cloud container: reported, not hidden. Gemma-4 cannot be served by this ROCm build (3 documented dead ends, incl. the w4a16 group-32 kernel gap on gfx1100): the session ran on Qwen3-14B. Carbon label: measured J/token x 571 g/kWh (China 2023, Ember).